

Title:

Bandpass Filter

Objective:

The objective of the bandpass filter is to retain EEG signals within a specific frequency range that contain meaningful brain activity, while removing unwanted low-frequency and high-frequency noise.

How it works:

Step 1: Split Data Trial by Trial

Step 2: How filter() work:

For example:

Original Signal: [2, 4, 6, 10, 6, 4, 2]

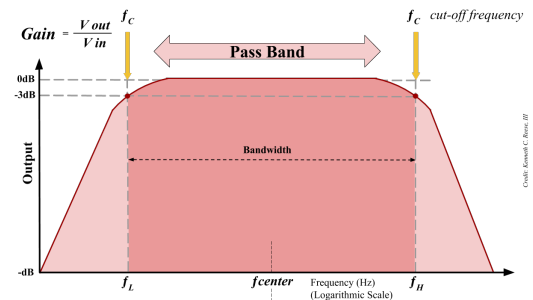
Window weights: [0.25, 0.5, 0.25]

new_value at position 3 = $0.25 \cdot 4 + 0.5 \cdot 6 + 0.25 \cdot 10$

new_value at position 3 = 6.5

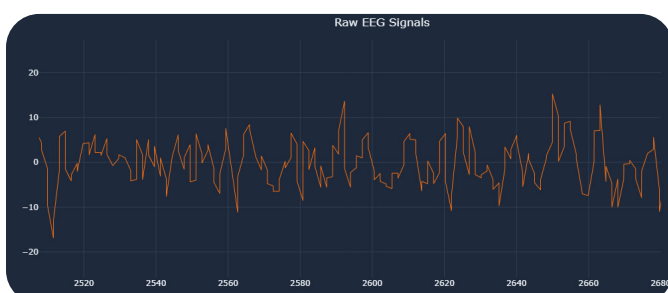
Step 3: Convert Numbers Back to DataFrame

Step 4: Recombine All Trials



- MNE's filter() works the same way but with bigger windows and exact weights calculated from the bandpass frequencies
- Every output number is a combination of its neighbors
- Forward-backward filtering

Before



After

